LIFELINE: Algorithmic Relief-Command Engine

Iteration 1 Approach Note: Data Generation, Planted Objects, Adversarial Traps & Computational Complexity

1. From Story to Numbers: Narrative Calibration & Parameter Choices

Delta Flood (Study 1)

Story: A flat delta of small villages joined by raised roads. Roads go under water in waves. Food and water matter most.

- Topological planar connectivity is modeled with average degree 2E/V in [2.2, 3.2].
- Flooding is modeled by over 40% of locations having high traversal penalties ≥ 2 (p in (2, 4, 6)) and at least 3 closure waves.
- Food (cat 0) and water (cat 1) crates account for ≥ 55% of inventory.

Mountain Earthquake (Study 2)

Story: A valley town with a few roads and almost no alternatives. The injured need medicine and tools. Bursts of urgent calls follow each aftershock.

- Alpine valley geography constrains network topology to a sparse backbone with almost no alternative cycles: average degree 2E/N in [2.0, 2.6].
- Physical trauma prioritizes medicine (cat 2) and rescue tools (cat 3) crates (≥ 45%).
- Aftershock bursts generate severe urgency spikes, requiring mean alarm priority ≥ 6.0.

Coastal Cyclone (Study 3)

Story: A port city and fishing villages. Storm surge closes roads in waves. Rescue windows are short and overlap. The radio is full of noise.

- Rapid inundation mandates evacuation refuge: ≥ 25% of locations are shelters (type S).
- Road closures occur across ≥ 3 waves.
- Tight tidal cycles produce short, heavily clustered rescue windows where every window

overlaps at least 2 other windows on average.
- Severe storm static uses ≥ 30 of the 38 radio alphabet symbols.

Urban Flash Flood (Study 4)

Story: A dense city with many alternative routes. Calls arrive faster than anyone can answer them. Rare equipment is available.

- Dense metropolitan grid provides high arterial redundancy: average degree $2E/V \geq 3.5$.
- Overwhelmed dispatchers experience ≥ 1.5 alarms per tick on tiers M and L.
- Municipal stockpiles provide specialized disaster technology, ensuring rare equipment (cat 4) comprises ≥ 10% of crates.

Landslide Corridor (Study 5)

Story: A single mountain highway along a gorge. A collapse cuts off whole communities.

- Canyon gorge restricts roads to a linear chain without bypasses: average degree ≤ 2.4 and no location has > 4 incident roads.
- Depth into the corridor requires unweighted distance to the farthest settlement to be ≥ floor(V/4) roads.
- Mass wasting occurs across ≥ 3 waves, and the union of all waves severs all paths to at least one target location from HQ.

Outbreak Camp (Study 6)

Story: A relief camp with an outbreak. Medicine and water count and supplies spoil. Radio beacons repeat the same messages.

- Infectious containment requires clean hydration and treatments: medicine and water crates (cat 2, 1) account for ≥ 55% of crates.
- Biological spoilage forces ≥ 50% of crates to expire by hour 300.
- Automated transmission repeaters rebroadcast static beacons, causing a single dominant symbol to occupy ≥ 25% of the total radio log.

2. Planted Objects & Claims: Exact Correctness by Construction

- **Planted Metric Manhattan Tour (TSP):** A closed permutation starting at site 0 is generated over all n coordinates in $[0, 999]^2$. Its length is evaluated as the closed Manhattan cycle sum. On Tiers L and X, the tour is constructed using the Nearest-Neighbor (NN)

heuristic starting from 0, guaranteeing length ≤ NN tour length by definition. On Tier S ($n \leq 9$) the global optimum tour_opt is evaluated via Held-Karp dynamic programming $O(n^2 \cdot 2^n)$.

- **Planted Vertex Cover (MVC):** A target subset C of size k in [2, floor(n/2)] is chosen. Every edge in sensor_edges is generated such that at least one endpoint is in C. All non-cover nodes are connected directly to C, and internal edges only connect pairs with at least one cover node. Thus, C is a valid vertex cover of size k by construction. On Tier S, the exact minimum cover size cover_min is evaluated via exact subset search.
- **Planted Safety Assignment (3-SAT):** A ground-truth truth assignment A is chosen for switches 1..m. Each 3-CNF clause is constructed by selecting one literal that evaluates to True under A (+s if A[s]=1, -s if A[s]=0). The other two literals are assigned random polarities. Consequently, every clause is guaranteed to be satisfied by A.
- **Planted Rations (Exact Subset Sum):** For m ration items with quantities v_i in [1, 1000], random binary selection flags b_i in (0, 1) are drawn. The target rations_target is set to sum(b_i * v_i). The planted subset sums exactly to target by construction.
- **Tier S Exact Claims:** For cases 00 to 19, exact algorithms solve 0/1 knapsack (cargo_opt), unweighted interval scheduling (window_count_opt), weighted interval scheduling (window_value_opt), rod cutting (drum_opt), metric TSP (tour_opt), and minimum vertex cover (cover_min) to provable global optimality.

3. The Four Trap Cases (Tier X: Cases 46 to 49)

Case & Name	Built Characteristics	Hostility / Targeted Failure	Validator Check
Case 46: Dock-Order Trap *(Study 5 Story)*	1,600 crates pre-sorted such that serial, expiry, weight, and value are monotonically non-decreasing.	Hostile to divide-and-conquer sorting routines. Unbalanced partitioning in Quicksort with first/last element pivot selection degrades to worst-case $O(n^2)$ time and risks	Validator check V8 verifies that in crate_id order, at least 3 of the 5 fields are monotone.

Case & Name	Built Characteristics	Hostility / Targeted Failure	Validator Check
		recursion stack exhaustion.	
Case 47: Best-Looking-Crate Trap *(Study 6 Story)*	12 crates and capacity W = 20 kg. One bait crate has weight 1 kg, value 12 (ratio 12.0); four optimal crates have weight 20 kg, value 100 (ratio 5.0); seven filler crates have weight 15 kg, value 10 (ratio 0.67). Total crate weight is 186 kg (W = 20 is 10.7%, in [10%, 70%]).	Traps greedy value-density heuristics. Sorting by value-density selects the 1 kg crate first, leaving 19 kg of capacity. None of the 20 kg crates can fit, forcing greedy to take a filler crate for total value 12 + 10 = 22. The true optimum selects a single 20 kg crate for value 100. Greedy achieves only 22% of optimal (≤ 60%).	Validator check V8 evaluates exact 0/1 knapsack DP vs greedy heuristic and verifies val_greedy ≤ 0.60 * val_opt.
Case 48: Shortest-Looks-Cheapest Trap *(Study 1 Story)*	1,600 locations containing a 550-hop linear spine (0-1-...-550) ensuring diameter ≥ 500 roads. Target 0 connects to HQ via a 1-hop direct highway with	Traps breadth-first search and hop-count heuristics. The path with the fewest roads has 1 road with cost 66, whereas the cheapest path has 2	Validator check V8 verifies network diameter ≥ 500 roads via BFS and verifies that for target 0, the minimum-cost BFS hop path is at least 1.5x the Dijkstra

Case & Name	Built Characteristics	Hostility / Targeted Failure	Validator Check
	traversal time 60 min and penalty 6 (cost 66), but an alternative 2-hop detour (0 -> 554 -> target_0) costs only 1 + 1 = 2 min.	roads with cost 2. Every minimum-hop route costs 33.0x more than the cheapest route (≥ 1.5x).	shortest path.
Case 49: Near-Miss Trap *(Study 2 Story)*	55,000 log symbols where 93% of characters are repetitions of 'A'. Patterns of length L in (2, 4, 8, 12, 20, 40) are structured with prefixes $'A'^{L-1}$.	Induces catastrophic backtracking in naive string matching algorithms. Every search position matches L-1 characters before mismatching on the terminal symbol, forcing O(N * M) worst-case comparisons in naive scan, whereas KMP processes it in strictly linear O(N + M) time.	Validator check V8 verifies block repetition ≥ 90% and confirms that for at least 4 patterns, the prefix of length L-1 matches at ≥ 50% of all start positions.

4. What Will Be Hard in Iteration 2: Bottlenecks & Algorithmic Hardness

Delta Flood

- **Route Planning (Module B):** High road penalties (≥ 40% with p ≥ 2) combined with

multi-wave dynamic closures require continuous re-routing and multi-criteria optimization between physical travel time and hazard exposure penalties.
- **Cable Network Planning (Module F):** Spanning tree construction must navigate waterlogged locations with severe cable weights across dynamically severed road networks.

Mountain Earthquake

- **Emergency Queue (Module D):** Alarm priorities are heavily compressed into the peak urgency band (mean priority ≥ 6.0). Under dense aftershocks, numerous incidents share identical priority, requiring multi-level tie-breaking and starvation prevention.
- **Relief Inventory (Module A):** High demand for heavy extraction tools and medical crates creates tight packing constraints where near-optimal knapsack solutions are critical.

Coastal Cyclone

- **Rescue Scheduling (Module C):** Severe interval contention where every rescue window overlaps at least two others precludes greedy choice, requiring weighted interval dynamic programming across hundreds of overlapping intervals.
- **Radio Logs (Module E):** Telemetry spans 30+ alphabet symbols with high atmospheric noise, making error-tolerant edit distance and prefix decoding computationally intensive.

Urban Flash Flood

- **Emergency Queue (Module D):** Influx rates exceeding 1.5 alarms per tick on tiers M and L generate large queue backlogs, making priority heap maintenance a throughput bottleneck.
- **Route Planning (Module B):** High topological density ($2E/V \geq 3.5$) expands the graph branching factor, significantly enlarging Dijkstra's priority queue frontier across Tier L networks (up to 2,000 nodes).

Landslide Corridor

- **Route Planning (Module B):** Network depth ($\geq \text{floor}(V/4)$ hops) along a single corridor means road closures cause complete community isolation, requiring rapid articulation-point detection and reachability boundary checks.
- **Network Planning (Module F):** Linear canyon topography forces spanning trees along fragile paths where single cable segment failures partition the entire downstream relief grid.

Outbreak Camp

- **Relief Inventory (Module A):** Over 50% of medical and water crates expire by hour 300.

Optimizing truck inventory requires multi-objective scheduling that simultaneously maximizes humanitarian utility and minimizes perishable spoilage loss.

- **Radio Logs (Module E):** Over 25% of log symbols are repetitive beacon characters, skewing frequency tables and requiring efficient run-length and Huffman tree compression to avoid memory bloat.